



Context-based information generation for managing UAV-acquired data using image captioning

Seongdeok Bang, Hyoungkwan Kim*

School of Civil and Environmental Engineering, Yonsei University, Republic of Korea

ARTICLE INFO

Keywords:

Image captioning
Text generation
On-site management
UAV

ABSTRACT

Unmanned aerial vehicles (UAVs) can efficiently collect image data representing various situations at construction sites, however, it requires a lot of time and cost to analyze it manually to retrieve useful information for on-site management. The paper proposes a methodology to generate time-spatial and visual context-based information from UAV-acquired data. This methodology generates textual information on the position, status, movement, color, and quantity of construction resources from site images using image captioning. Then, construction site images, text generated from them, and UAV flight data containing the latitude, longitude, date, and time of day, are systemized into a database. For evaluating the proposed methodology, data obtained by UAV at actual construction sites was used. Our methodology could predict textual information with a mean average precision of 43.52%, which is superior to those of existing methods.

1. Introduction

The development of data-collection and storage technologies allows us to collect enormous amounts of data and manage them with a computerized system. In this context, data generated during the life cycle of a construction project, such as planning, design, construction, and operation, is traditionally stored in the form of documents, images, web pages, and Building Information Modeling (BIM) [1–3]. As technologies such as sensor networks and the Internet of Things (IoT) are being commercialized, the amount of data available in the construction industry is expected to increase even more. Based on the “big data” accumulated in the construction industry, many attempts have been made to extract useful knowledge to support construction project management. To develop an effective project management system, it is necessary to consider the surrounding environmental, temporal, and spatial context when extracting information from the available data.

The introduction of unmanned aerial vehicles (UAVs) has enabled image data collection to represent various situations in a wide range of construction sites. UAVs can generate a large number of construction site images in a short time owing to their excellent maneuverability. UAV-acquired construction site images provide managers with visual information, such as the presence of construction resources and progress of construction work. From these images, managers can recognize construction resources and understand the construction process occurring on a site. However, considerable time and effort are required to

analyze a large number of construction site images manually. When a construction site is large and the data-recording period becomes longer, this work becomes more difficult. Advancements in computer-vision technology make it possible to obtain useful information from images automatically. In particular, image captioning, a computer vision-based technology, is gaining traction in this field [15–17]. This technology understands the characteristics of objects, their activities, and relationships and expresses them in the natural language. The use of image captioning may enable efficient management of a large number of construction site images acquired per day by enabling the administrator to search the image they want to check in a short time.

Previously, image captioning studies have been conducted on datasets consisting of everyday images, such as the PASCAL Sentence dataset [18], Flickr 30K [19], Microsoft COCO dataset [20], and Visual Genome [21]. However, site managers need more high-level managerial information than these techniques can provide. This information includes knowledge created based on the contextual interpretation of objects and situational awareness from images, such as the quantity of materials, installation status of specific structures, and the movement of workers or construction equipment. Based on image captioning technology, studies should be conducted to extract such information from construction site images. However, because image captioning only extracts information from images, other information that is difficult to obtain from images should be acquired separately. At construction sites, similar images may be obtained even if they were taken at different

* Corresponding author.

E-mail addresses: bangdeok@yonsei.ac.kr (S. Bang), hyoungkwan@yonsei.ac.kr (H. Kim).

<https://doi.org/10.1016/j.autcon.2020.103116>

Received 10 July 2019; Received in revised form 24 January 2020; Accepted 1 February 2020

Available online 07 February 2020

0926-5805/ © 2020 Elsevier B.V. All rights reserved.

times and locations because of the repetition of certain tasks in some processes. Thus, time and location information should accompany the captured image. The textual information extracted from images, such as location, color, states, movements and quantities of key construction resources, can generate new information by combining with time and space data. For example, this information can be used as evidence of litigation between project stockholders or schedule management of construction processes by comparison with as-planned data. It can also help make daily logs, which periodically record as-built data of construction sites.

UAVs can replace manual monitoring at a large-scale construction sites by obtaining visual data efficiently. To effectively manage the large amount of data acquired from UAVs, we propose a methodology to generate context-based information from construction site images using UAV-acquired data and image captioning. The proposed methodology predicts sentences to describe images using dense captioning, which is a subset of the image captioning technology. The produced sentences provide information on the position, status, movement, color, and quantity of construction resources. Then, unstructured data, including construction site images, text generated by the first technique, and flight data, including the time and location corresponding to each image are systemized, thus resulting in the creation of database. This database provides spatiotemporal information with textual description to site managers. Experimental results demonstrate that the proposed methodology can effectively generate context-based information from UAV-acquired data.

2. Related studies

2.1. Information retrieval from construction site images

Images taken at construction sites have been widely used to retrieve on-site information for construction management. Some studies have tried to address the safety problem by determining individual workers' safety level or identifying whether workers are wearing safety equipment. Kim et al. [4] suggested a proximity-measurement method in construction resources using 2D images collected from a UAV and YOLO-v3 model. Kim et al. [22] proposed a vision-based hazard-avoidance system to augment the hazard information of each construction worker on a wearable device using fuzzy-based reasoning of the safety level. Kolar et al. [23] developed a detection model for safety guardrails to prevent workers from falling from heights using transfer learning and the VGG-16 model. Fang et al. [24] developed Faster R-CNN and deep CNN models to detect workers and their harnesses for proactively identifying unsafe behavior and mitigating the likelihood of falls from heights. Mneymneh et al. [25] proposed a vision-based monitoring framework for hardhat wearing by developing motion-based human detection and color-based hardhat detection.

Meanwhile, some researchers focused on automatically monitoring the progress of construction work. Bang et al. [9] suggested the automatic generation of high-resolution maps to easily identify the status of a construction site using UAV-acquired images and image stitching. Han and Golparvar-Fard [26] presented an appearance-based material recognition method using 4D BIM and 3D point cloud models for detecting operation-level progress. Kropp et al. [27] suggested an indoor progress-monitoring method to compare the current state of construction activities from as-built video with as-planned 4D BIM information. Hamledari et al. [28] identified the components of an interior partition and classified them into five states (framing, insulation, installed dry-wall, plastered drywall, and painted partition) for tracking progress. Turkan et al. [29] proposed a tracking system for temporary and secondary components, such as formwork, scaffolding, shoring, and rebar, to estimate construction progress using photogrammetry and 3D terrestrial laser scanning.

Other studies analyzed construction site images for evaluating productivity in a construction process. Gong and Caldas [30] proposed

a vision-based video-interpretation model to generate productivity information with a case study on the concrete column pour process. Kim et al. [31] presented an algorithm for analyzing earthmoving productivity to automatically generate site-access documentation based on the detection of license plates of construction vehicles. Bögler et al. [32] suggested a productivity-estimation method for soil removal by quantifying the volume of excavated soil via photogrammetry and video analysis. Kim et al. [33] analyzed the productivity of a tunnel earthmoving process based on WebCYCLONE simulation and detection of excavators and dump trucks. Kim and Chi [34] proposed a framework for recognizing the movement of excavators to compute their earthmoving cycle time.

All the studies cited above employed construction resource recognition for contextual reasoning in images. For example, in safety assessment studies, hazardous situations were defined using statistical information, such as the distance between a worker and equipment, based on the detected resources. To obtain contextual information, they combined computer vision technologies with inference methods such as fuzzy logic, rule-based expert systems, and appearance-based approaches. These inference methods should be integrated with computer-vision technology to develop more automatic and meaningful generation of contextual information from images.

2.2. Image captioning technology

Image captioning is a technology that automatically generates textual descriptions from images for semantic indexing and retrieval from large image databases or extracting information or insights. This technology performs object recognition from an image, infers the relationship between the recognized object to understand the event in the image, and generates an analysis in the natural language. To this end, image captioning was initially studied to obtain image features such as its color, texture, and descriptor and to match them with a set of words using a training dataset with a given set of annotation words for each image [35–38].

Deep learning-based image captioning methods extract feature vectors from images and translate them into the natural language using both natural language-processing (NLP) and computer-vision technologies. Recently, a deep neural network, composed of a convolutional neural network (CNN) followed by a recurrent neural network (RNN) language model to implement NLP and computer vision in one model, has been designed [16,39,40]. In this design, two approaches, top-down and bottom-up, were employed. The top-down approach obtains image features, such as color, texture, and descriptor, and translates them into sentences using a set of words [35,39]. This approach finds it difficult to deliver detailed descriptions of an image as it uses feature vectors extracted from the entire image. This disadvantage becomes more pronounced in the analysis of construction site images because they have more complex backgrounds and more objects than the images of such dataset used in the computer-vision field. In contrast, the bottom-up approach derives words from specific regions of an image and combines them for predicting sentences. To emulate the way in which humans perceive an image, it is necessary to focus on the important parts of the image rather than uniformly analyzing the entire image. However, the bottom-up approach simply combines keywords extracted from an image to produce misleading results that do not make sense.

Dense captioning [15] was proposed to generate dense and rich annotations by image captioning in several regions of an image using both top-down and bottom-up approaches. Dense captioning localizes regions of interest and describes each of them in the natural language by performing object-detection, soft spatial-attention, and image captioning tasks simultaneously in one model. The process of dense captioning is as follows. First, object candidate regions are produced by applying soft spatial attention to the region proposal network (RPN) of the faster region-based convolutional neural network (R-CNN) [41]. Soft spatial attention enables the smooth extraction of a fixed-size

vector from regions of different sizes and orientations because it can process arbitrary affine transformed regions. Later, the fixed-size vector extracted from an image is input to a RNN language model, which is trained to predict the next word based on a vector extracted from the image and a sequence of previously used words. Word prediction is repeated until RNN determines that all words have been predicted from the vector, and the sequence of predicted words constitutes a sentence as the output of the RNN model for each image region.

Unlike other image captioning methods, dense captioning is useful for analyzing construction site images with many objects and multiple relationships; it focuses attention on image regions containing objects. However, it is difficult to provide the managerial information required by construction managers using expressions produced by the existing methods (e.g. “skateboard with red wheels” or “a cat riding a skateboard”). To address this problem, it is necessary to define textual descriptions useful for on-site management and to create them. Thus, this study aims to develop a construction information generation method to express detailed events occurring on construction sites in a text format.

3. Methodology

The methodology for context-based construction information builds a database considering time-spatial context and situational awareness at a construction site from UAV-acquired data consisting of images and flight data. First, construction resources are detected from an image and sentences are generated describing their position, status, movement, color, and quantity using the dense captioning technology. Next, the latitude, longitude, and date and time of day corresponding to the image are imported from UAV flight data. Finally, data with location, time, and generated textual descriptions for each construction site image is structured into a database. Thus, time-spatial and visual context-based information is retrieved from UAV-acquired data. In this study, the context-based information can be generated based on automatic situational awareness with considering the surrounding environment by using analysis results of data collected from sensors attached to UAVs. This information can visually represent specific construction sections at a given time with textual descriptions of construction resources. Fig. 1 shows the framework of the proposed methodology.

UAV-acquired construction site images provide information on the actions of construction resources, their state, identification, quantity, and relationship with other objects or the background. This information can be used as a basis for on-site management practices, such as

proactively preventing accidents by assessing the safety level of construction workers, analyzing the productivity of earthworks by recognizing the amount of soil in dump trucks and movement of excavators, and monitoring work progress by identifying the number of key materials and installed temporary structures. To automatically extract such information, the textual description technique makes use of dense captioning. Fig. 2 shows a brief schema of the dense captioning used in this technique. The input image is first processed by Faster R-CNN to output object-candidate region vectors. Region vectors are produced using the soft attention mechanism implemented in a localization layer. Region vectors are then entered into the long-short term memory model (LSTM) to produce their own sentences. The LSTM, a type of RNN, recognizes patterns in sequence-type data; it was developed to solve the problem of long-term dependence in which previous information is not sufficiently transmitted when the time-step of the RNN increases. The LSTM employs a forget gate to improve the accuracy of sentence predictions by applying a low weight to non-critical information and a high weight to critical information. When the LSTM determines the end of a sentence, it stops the prediction process and outputs the sequence of words predicted to that point as a sentence.

In this study, five types of information, including position, status, movement, color, and quantity, were defined for generating the managerial information required by construction managers. Firstly, position provides the background information on construction resources. Depending on site characteristics, construction resources can be located on natural substances, such as soil, rocks, and grass, or on structures built during the construction process, such as concrete, steel plates, and scaffolds. The position information is the basis for identifying and tracking the location of key material or equipment. Secondly, status provides information on the state of construction resources. For example, information on whether structures are installed or not, whether materials are covered by soil or other resources, and whether a construction equipment is loaded with something or is empty is included in the status information. The status of construction resources is understood in relation to the surrounding environment. Movement describes the behavior of portable construction resources. The information specifies the operation type of resources to identify various construction works. For example, the operation types of the excavator can be identified by swing, digging, or hauling. Meanwhile, color provides the appearance information for accurate identification of construction resources. When similar construction resources performing different tasks are on a site, they can be better classified with the color information for clear situational awareness. Finally, quantity recognizes the number or

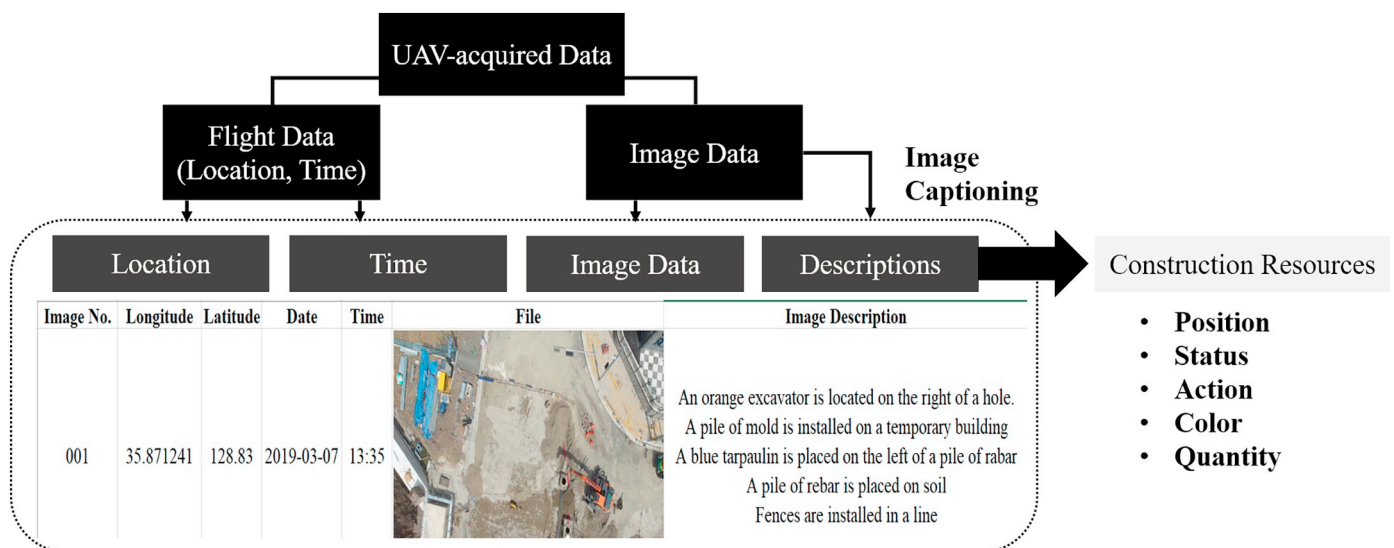


Fig. 1. Framework for generation of context-based construction information.

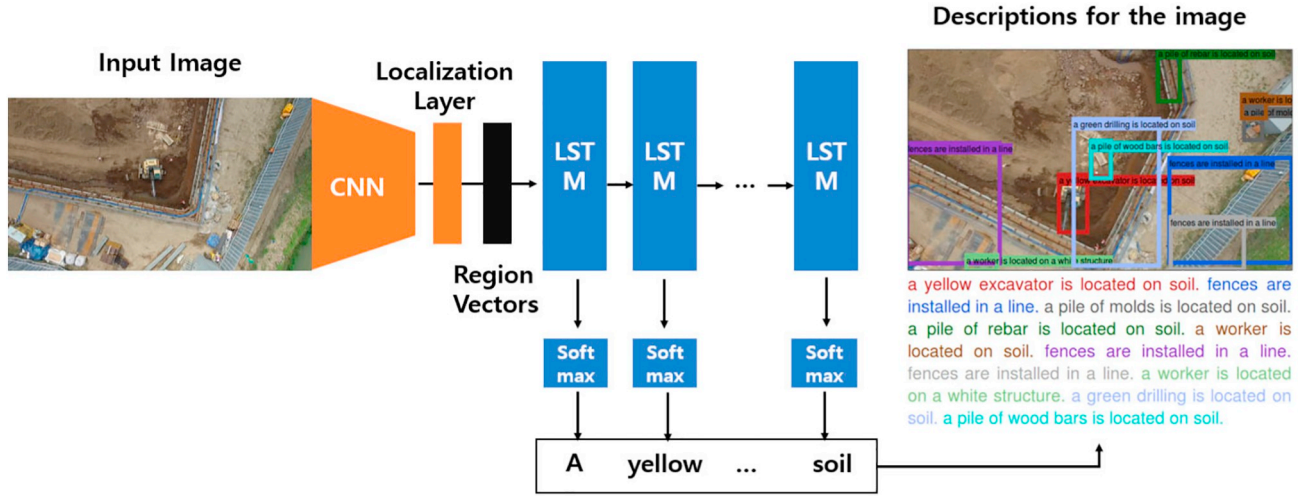


Fig. 2. Brief schema of the dense captioning model.

Table 1
Example sentences created based on the five types of information.

Information type	Examples of sentences
Position	A pile of rebar is located on soil A concrete pillar is installed on concrete
Status	A pile of mold is installed on a temporary building A silver dump truck is loaded with soil
Movement	An orange excavator is putting soil into a dump truck An orange excavator is digging soil
Color	A blue dump truck is located on soil An orange excavator is located on soil
Quantity	Three piles of rebar are located on concrete Twelve concrete pipes are located on soil

A bold font indicates the information type that examples of sentences want to provide.

amount of construction resources. Stacked materials that are difficult to count accurately in an image are quantified in a pile unit. This information can be used to compare between as-planned and as-built. Table 1 shows example sentences generated using the five types of information described above. The dense captioning model was trained to predict sentences in the manner described in Table 1 from the input image. As the output of this technique, textual descriptions are generated from construction site images using the dense captioning model.

4. Experimental studies

4.1. Experimental environment and the dataset

The proposed methodology was implemented on an Ubuntu 16.04 operating system with an Intel Core i7-7700 processor and GTX 1080 Ti GPU, using Python and Lua programming languages. Torch, a machine-learning library, was used to train and test the dense captioning model. Microsoft Excel was used to systemize UAV-acquired data on a spreadsheet. A Phantom 3 Professional UAV (DJI Corporation) was used to collect construction site images. The dataset consisted of 1431 images obtained from six different construction sites including 8428 captions generated using the textual description technique; each caption corresponded to a region in a given image. As shown in Fig. 3, the images and captions in the dataset were manually annotated using VGG image annotator [42]. The resolution of the original image was 3840×2160 ; however, for computational efficiency, it was adjusted to 960×540 after preserving the aspect ratio of the image. The dense captioning model was trained with a learning rate of 0.000001. An Adam optimizer was used with exponential decay rates for the first and

second moments of 0.900 and 0.999, respectively. The number of iterations and dropout probability in the training process were set to 100,000 and 0.5, respectively.

4.2. Pre-processing on the dataset

Five pre-processing techniques were applied to the dataset for improving the performance of text prediction from construction site images. Two pre-processing techniques were applied to bounding boxes indicating the object regions of images and the other three pre-processing techniques were applied to captions describing these object regions. The first pre-processing merged two bounding boxes with overlapping areas of more than 80% into one box with multiple captions. The two overlapping boxes were merged by taking the average values of each position information (x- and y-coordinates of the centre point, width, and length). This technique prevents confusion in the object recognition process by integrating the position information of different bounding boxes representing the same object into one. The second pre-processing removed a bounding box whose area was less than 2500 pixels. Because our datasets consist of images taken by UAVs at high altitudes, they include many small objects. These small objects do not have sufficient evidence (i.e. visual features in the images) for the model to extract their information. Thus, we excluded areas and captions of images for small objects from the training and testing process.

Natural language data typically includes a character string that prevents data from being interpreted, such as misspelled words or special characters. To learn a meaningful pattern from text data, a pre-processing technique is essential to eliminate these strings and simplify the predictive scope of the language model, which should predict the probability of a word sequence. To this end, the three pre-processing techniques, which were applied to captions of the dataset, aim to reduce the dimension of the dictionary. A dictionary is a set of words used to predict sentences, which works by encoding all words in the training data into numbers that the language model can identify. A high dimension in the dictionary, i.e. a large number of words used to predict sentences, not only reduces sentence-prediction accuracy in the language model but also increases the calculation time.

The first pre-processing for captions replaced all uppercase letters in the dataset with lowercase letters. In the dataset, the first letter of a sentence usually begins with an uppercase letter. Sometimes, words that should be expressed in lowercase letters may be misrepresented with uppercase letters. This pre-processing is necessary because the dimension of the dictionary is increased by recognizing upper and lowercase letters separately in the programming language. The second

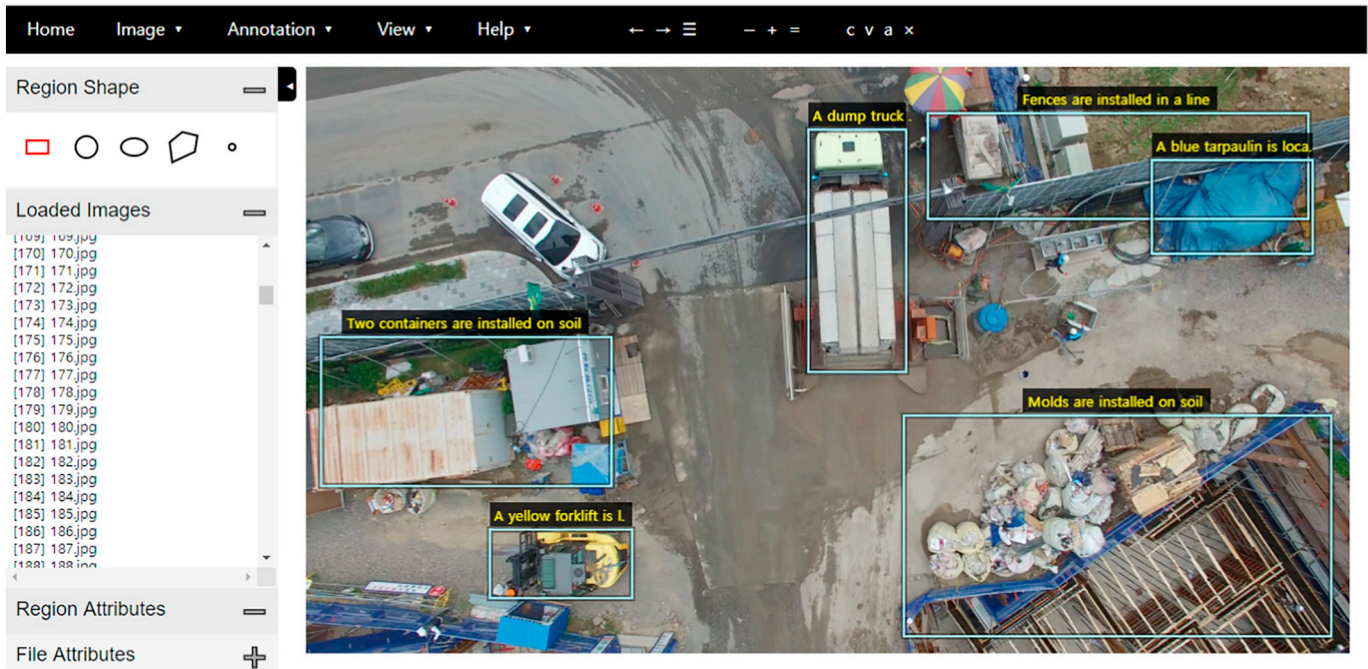


Fig. 3. Example of manual labelling using the VGG image annotator with image regions and captions.

Table 2

Types of words with a frequency of five or less in the dataset and their examples.

Type	Words
Unusual quantity information	Fifteen, eleven, sixteen, nineteen, twentyone, thirtytwo, twentytwo, twentythree, twentyseven
Unusual color information	Brown
Typing error	Loaded, plie, exca, builing, os, soi
Spacing error	Installedtwenty, locatedon, soila

pre-processing was to perform Lemmatization, which converts words into a root word. For example, the words “changing” and “changes” are converted into the word “change.” Lemmatization was performed using the WordNetLemmatizer function of natural language toolkit (NLTK), one of Python's packages. The last pre-processing was to remove words with a frequency of five or less from the dataset. Table 2 shows the types of removed words from the dictionary and their examples. Words with a low frequency in the dataset include misspellings and incorrect spacings, which occur during the manual process of creating captions. These words must be removed because they prevent the language model from undergoing correct training. In addition, low-frequency

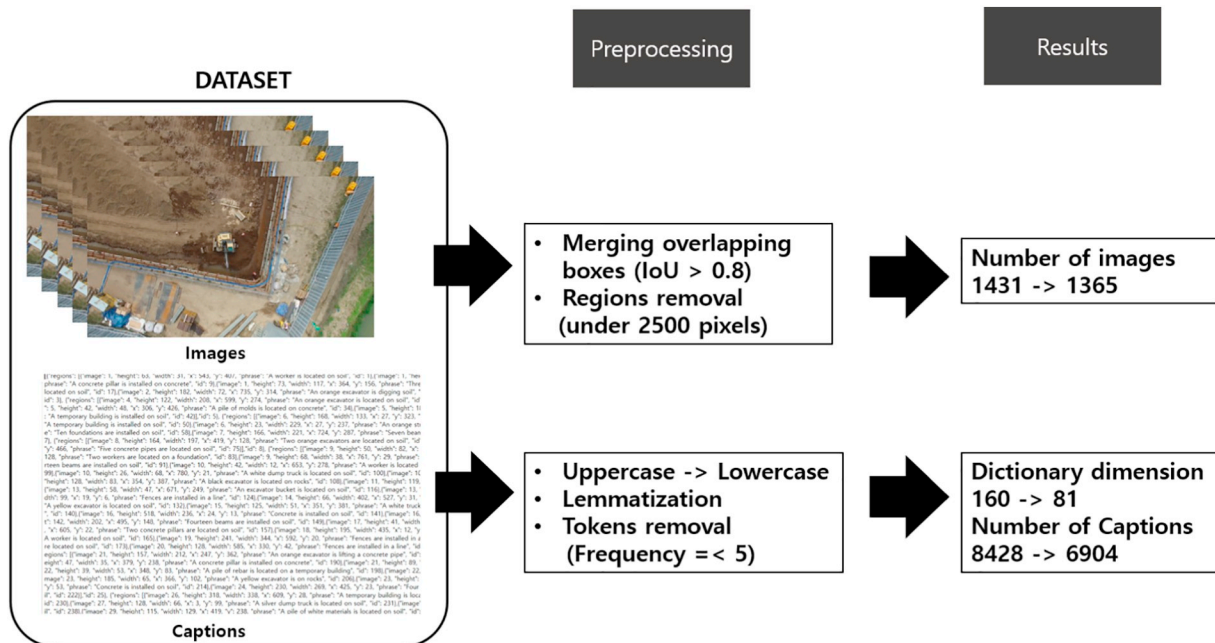


Fig. 4. Application of the five pre-processing techniques to our dataset.

Image No.	Longitude	Latitude	Date	Time	Image Description	File
001	35.8712	128.83477	2019.03.07	13:35	An orange excavator is located on the right of a hole. A pile of mold is installed on a temporary building A blue tarpaulin is placed on the left of a pile of rabar A pile of rebar is placed on soil Fences are installed in a line	
002	35.8712	128.83478	2019.03.08	15:34	An orange excavator is located on the right of a hole. A pile of mold is installed on a temporary building A blue tarpaulin is placed on the left of a pile of rabar A pile of rebar is placed on soil Fences are installed in a line A pile of Rebar is placed on soil	

Fig. 5. Comparison between (a) ground-truth and (b) predictive result. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

words represent unusual quantities and color information in the dataset. Image areas and captions with low-frequency words were excluded from the experiment because they were determined to be lacking in training data. Fig. 4 schematizes the results of the application of the five described pre-processing techniques on our dataset. As a result of pre-processing, the number of images, number of captions, and dimensions of the dictionary were reduced from 1431 to 1365, 8428 to 6904, and 160 to 81, respectively.

4.3. Results

Fig. 5 shows an example of comparing the ground-truths to the results produced by using the textual generation technique. Fig. 5(a) shows the ground-truth for image regions and captions of an image chosen from 365 test images and Fig. 5(b) shows the prediction results of region detection and region-specific captioning. All seven objects in Fig. 5(b) were detected by 10 bounding boxes, but the proposed methodology fails to predict all ground-truth sentences in Fig. 5(a). The bounding box marked in blue in Fig. 5(a) shows that the truck has ground-truth information on the position, status, quantity and color. However, Fig. 5(b) shows that the proposed methodology fails to generate status information, which is “a white truck is empty.” Although errors in object recognition or sentence prediction occur as shown in Fig. 5(b), the proposed methodology succeeds in producing six out of eight information.

Fig. 6 shows an example in which data with latitude, longitude, date, time of day, construction site images, and generated text information are organized into a spreadsheet format. Image files stored in a computer were entered into the cells of the spreadsheet using the image and table tag of the HTML program (Eq. (1)).

$$\langle \text{img src} = \text{"Directory"} \text{width} = W \text{ height} = H \rangle \langle \text{table} \rangle \quad (1)$$

Here, *Directory* indicates the path of the image file including the file name in a computer, and *W* and *H* denote the width and height, respectively, based on which the image will be inserted into the cell of the spreadsheet file. The process to build a database using a spreadsheet file is as follows. First, to import the image files in numerical order, their names are set to the index numbers shown in the first column of Fig. 6.

Then, *Directory* is partly represented by using a cell address with an index number equal to an image file name (e.g., “C:\images\“&A2&”.jpg”). With the autofill function of the spreadsheet, the images are automatically inserted into the seventh column of Fig. 6. Finally, the proposed methodology imports flight data corresponding to the image. Phantom 3, which is the UAV employed for collecting data, recorded 32 types of flight information, including latitude, longitude, and altitude every 0.1 s from the time of taking off until landing. Among these flight information data, there is a binary value called *isPhoto*; it records a value of 1 when the UAV takes an image. The methodology takes the information (latitude, longitude, date, and time of day) with an *isPhoto* value of 1 from UAV flight data and inputs them into a spreadsheet file as shown in Fig. 6.

4.4. Evaluation and discussion

To demonstrate the proposed methodology, the dense captioning model was trained using the dataset with the five pre-processing techniques. Of the 1365 images in the dataset after pre-processing, 1000 images were used to train the dense captioning model and the remaining 365 images were used to test the model (Fig. 7). The dataset was randomly and exclusively divided into training and test dataset. Some of the training and testing images may be obtained from the same construction site, however, they are acquired at different times and are not visually identical. The predicted results were measured in terms of the mean Average Precision (mAP), which have been used in previous dense captioning studies to evaluate localization and language similarity simultaneously [15,17]. Localization accuracy was determined by intersection of union (IoU), Fig. 6. Formation of a spreadsheet by the data-organization technique which represents the quality of match between the prediction bounding box and the reference. Meanwhile, language prediction accuracy was determined using Metric for Evaluation of Translation with Explicit Ordering (METEOR), which yields the harmonic mean of precision and recall. METEOR was used in this study as it had been considered the most relevant indicator in image description evaluation [43]. Mean Average Precision was calculated by averaging at different IoU thresholds of 0.3, 0.4, 0.5, 0.6, and 0.7 and METEOR thresholds of 0, 0.05, 0.1, 0.15, 0.2, and 0.25. These

(a) Ground-Truth image



A red structure is located on soil

A container is installed on soil

A white truck is located on soil

A blue truck is located on soil

A white truck is located on soil

A white truck is empty

An orange excavator is located on soil

Two safety cones are located on soil

(b) Resultant image



a pile of beam are located on soil

two container is installed on soil

a white truck is located on soil

a blue truck is located on soil

a white truck is located on soil

an orange excavator is located on soil

four safety cone are located on soil

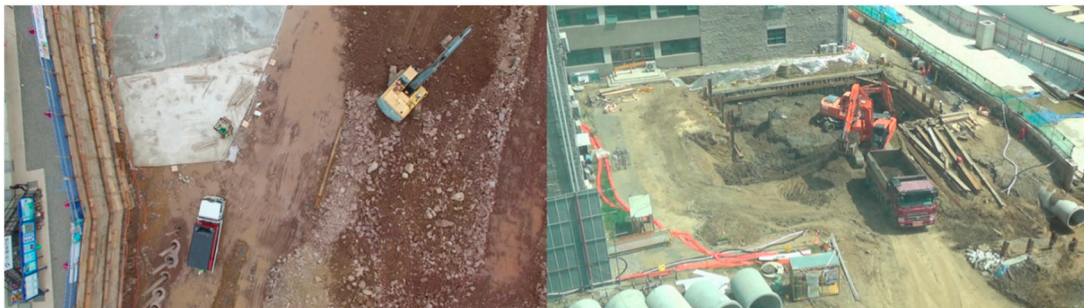
two container are installed on soil

a white truck is located is located on soil

a blue truck is located on soil

Fig. 6. Formation of a spreadsheet.

(a)



(b)



Fig. 7. Examples of (a) training image and (b) testing image.

(a) With preprocessing



a white truck is located on soil
a red crane truck is located on soil
four beam are located on soil
four beam are located on soil
eleven beam are installed on soil
four beam are located on soil
two pile of wood bar are located on soil
eleven beam are installed on soil
a pile of rebar is located on soil
four beam are located on soil

(b) Without preprocessing



A white truck is located on soil
A worker is located on soil
An orange excavator is located on soil
A worker is located on soil
Fences are installed in a line
A pile of rebar is located on soil
A pile of wood bars is located on soil
A worker is located on soil
A worker is located on soil
A worker is located on soil
Four beams are located on soil

Fig. 8. Comparison between prediction results: (a) image with pre-processing and (b) image without pre-processing.

thresholds were the criteria for determining the predictive success of localization and language prediction. For example, if the METEOR between the predictive sentence and the ground-truth was greater than or equal to the threshold, the language prediction was considered to be successful. To compare the performance of the proposed methodology with the existing dense captioning studies [15,17], the thresholds for IoU and METEOR used in this study were applied in the same way as the previous studies. The experimental results confirmed that it is feasible to generate textual descriptions from construction site images using the dense captioning model using the five types of information on construction resources. Although we experimented with a dataset different from that used for existing studies, Johnson et al. [15] and Yang et al. [17] showed the mAP of 5.39% and 9.31% for the visual genome V1.0 dataset, respectively.

Table 3 shows the differences in the performance of the models according to whether the pre-processing was performed or not. Our

Table 3

Comparison of image-description performance (mAP) of the proposed methodology with and without pre-processing.

Model	mAP	Number of training and test images
Without pre-processing	37.63%	Training 1066, Test 365
With pre-processing	43.52%	Training 1000, Test 365

Table 4

Mean average precision (mAP) of text generation by information type and number of captions in the test dataset.

Information type	mAP	Number of captions
Position	43.50	1873
Status	41.27	653
Movement	2.73	143
Color	47.72	592
Quantity	26.18	592
Total	43.52	2103

experimental results demonstrated that the proposed pre-processing techniques were effective in improving the performance of the model by increasing the mAP by 5.89%. Referring to Fig. 8(b), the model without pre-processing misjudged six small-size image regions as the existence of construction workers and predicted sentences based on the misidentified construction resources. However, referring to Fig. 8(a), the model with pre-processing could generate more accurate text information because it resulted in relatively few object detection errors. The predictive sentences in Fig. 8(a) did not start with a capital letter and included a subject-verb agreement error (e.g. “eleven beam are” not “eleven beams are”) due to the pre-processing, but these errors do not matter in information delivery.

Table 4 shows the accuracy of text generation by information type

and the number of captions in the test dataset. The accuracy corresponding to information type (position, status, and color) was similar to the average mAP. However, the accuracy corresponding to quantity and movement information was significantly lower than the average mAP. In particular, the mAP corresponding to movement was 2.73%, showing that the proposed method rarely yields information that can describe the movement of construction resources in the image. These results highlight the limitations of the proposed methodology. This methodology finds it difficult to clearly understand physical movements because it extracts information from a single static image. Furthermore, quantity information cannot be generated correctly unless the model accurately detects all the construction resources in the image.

Here, we present two suggestions to improve the proposed methodology. First, video data should be analyzed to understand the movement of construction resources. Physical measurements of objects, such as their acceleration, speed, and direction, can be understood using object tracking algorithms based on video analysis. This analysis can provide site managers with details of the actions of portable construction resources in the form of textual description. For example, it is possible to measure the safety level of an individual worker or equipment performance such as idle time and work efficiency, and deliver them in text format. Secondly, new models and training data, which can improve the accuracy of object detection, should be developed. Deep learning-based object detection studies have been conducted in the field of construction; however, their generalization performance is not yet high enough for a variety of actual construction sites. The development of new object detection algorithms, customized to suit the characteristics of construction sites with considerations for domain knowledge, can improve the performance of this methodology. Such improvements can be strengthened when new algorithms are trained with large amounts of image data reflecting various situations at construction sites.

5. Conclusion

This study proposes a methodology for generating context-based information from UAV-acquired data. The methodology provides information on the position, status, movement, color, and quantity of construction resources from UAV-acquired construction site images. Deep learning-based dense captioning creates textual descriptions from the regions of an image based on object detection. Then, the methodology constructs a database in the form of a spreadsheet, which consists of construction site images, time and location from UAV flight data, and text information generated by the textual description technique. To validate the proposed methodology, a dataset consisting of 1431 images captured by a UAV at six different construction sites and 8601 captions describing the regions of the images was utilized. Of 1365 images processed by the five pre-processing techniques, 1000 images were used to train the dense captioning model and the remaining 365 images were used to test the model. The dense captioning model predicted sentences from the test images describing the construction site with a mAP of 43.52%, which showed a higher performance than those of existing dense captioning studies. The predicted sentences were used to extract context-based information from UAV-acquired construction site images along with time and location information to build a database. The experimental results demonstrated that the proposed methodology can effectively generate context-based information from UAV-acquired data.

The contribution of this study to the construction industry is two-fold. First, our study presents an image captioning-based methodology to generate time-spatial and visual context-based information from UAV-acquired construction site images. To effectively manage the enormous amount of UAV-acquired data at construction sites, the proposed methodology systemized the database with attributes such as construction site image, text-based image description, and location and time of construction site, in order to provide site managers with

managerial information. Using the generated context-based information, the managers can identify the amount and exact location of specific construction resources, and the installation status of the temporary structure in the particular construction section at a given time using the database generated by the proposed methodology. Furthermore, based on the context-based information, if the construction progress for a particular section is later than the planned schedule, the managers may make decisions such as increasing the amount of resources input into the section or changing the order of the construction process. Our experimental results showed the feasibility of deep learning-based image captioning in the construction industry. Second, a dataset consisting of construction site images and captions was created. Existing image datasets in the construction industry include the identification and position information of construction resources for classification, detection, and segmentation. In contrast, our dataset contains five types of text information on construction resources, as well as their identification and position information. The dataset can continue to be used as a benchmark for textual information generation from construction site images.

The purpose of this study is to develop a methodology for effectively managing the enormous amount of construction site data obtained by UAV. The results of this study are expected to increase the usability of UAV-acquired data by generating context-based construction information and further contribute to the automation of construction. However, future research is required to apply this methodology to real construction sites. The difficulty in creating movement information using a static image and the low performance with respect to construction-resource recognition would suggest the direction of future research efforts.

Acknowledgements

This work was supported by the Graduate School of YONSEI University Research Scholarship Grants in 2019 and a National Research Foundation of Korea (NRF) grant funded by the Ministry of Science and ICT (No. 2018R1A2B2008600) and the Ministry of Education (No. 2018R1A6A1A08025348).

References

- [1] A.H. Alavi, A.H. Gandomi, Big data in civil engineering, *Autom. Constr.* 79 (2017) 1–2, <https://doi.org/10.1016/j.autcon.2016.12.008>.
- [2] M. Bilal, L.O. Oyedele, J. Qadir, K. Munir, S.O. Ajayi, O.O. Akinade, H.A. Owolabi, H.A. Alaka, M. Pasha, Big data in the construction industry: a review of present status, opportunities, and future trends, *Adv. Eng. Inform.* 30 (3) (2016) 500–521, <https://doi.org/10.1016/j.aei.2016.07.001>.
- [3] K.K. Han, M. Golparvar-Fard, Potential of big visual data and building information modeling for construction performance analytics: an exploratory study, *Autom. Constr.* 73 (2017) 184–198, <https://doi.org/10.1016/j.autcon.2016.11.004>.
- [4] D. Kim, M. Liu, S. Lee, V.R. Kamat, Remote proximity monitoring between mobile construction resources using camera-mounted UAVs, *Autom. Constr.* 99 (2019) 168–182, <https://doi.org/10.1016/j.autcon.2018.12.014>.
- [5] S. Bang, H. Kim, H. Kim, UAV-based automatic generation of high-resolution panorama at a construction site with a focus on preprocessing for image stitching, *Autom. Constr.* 84 (2017) 70–80, <https://doi.org/10.1016/j.autcon.2017.08.031>.
- [6] J. Johnson, A. Karpathy, L. Fei-Fei, Densecap: fully convolutional localization networks for dense captioning, *Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2016)*, Las Vegas, NV, pp. 4565–4574, doi:<https://doi.org/10.1109/CVPR.2016.494>.
- [7] A. Karpathy, L. Fei-Fei, Deep visual-semantic alignments for generating image descriptions, *Proceedings of the 2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2015)*, Boston, MA, pp. 3128–3137, doi:<https://doi.org/10.1109/CVPR.2015.7298932>.
- [8] L. Yang, K. Tang, J. Yang, L.-J. Li, Dense Captioning With Joint Inference and Visual Context, *arXiv preprint arXiv:1611.06949*, (2016).
- [9] M. Everingham, S.A. Eslami, L. Van Gool, C.K. Williams, J. Winn, A. Zisserman, The pascal visual object classes challenge: a retrospective, *Int. J. Comput. Vis.* 111 (1) (2015) 98–136, <https://doi.org/10.1007/s11263-009-0275-4>.
- [10] B.A. Plummer, L. Wang, C.M. Cervantes, J.C. Caicedo, J. Hockenmaier, S. Lazebnik, Flickr30k entities: collecting region-to-phrase correspondences for richer image-to-sentence models, *Proceedings of the 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2016)*, Las Vegas, NV, pp. 2641–2649, doi:<https://doi.org/10.1109/ICCV.2015.303>.

- [20] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, C.L. Zitnick, Microsoft Coco: Common Objects in Context, *arXiv preprint arXiv:1405.0312* (2014).
- [21] R. Krishna, Y. Zhu, O. Groth, J. Johnson, K. Hata, J. Kravitz, S. Chen, Y. Kalantidis, L.-J. Li, D.A. Shamma, Visual genome: connecting language and vision using crowdsourced dense image annotations, *Int. J. Comput. Vis.* 123 (1) (2017) 32–73, <https://doi.org/10.1007/s11263-016-0981-7>.
- [22] K. Kim, H. Kim, H. Kim, Image-based construction hazard avoidance system using augmented reality in wearable device, *Autom. Constr.* 83 (2017) 390–403, <https://doi.org/10.1016/j.autcon.2017.06.014>.
- [23] Z. Kolar, H. Chen, X. Luo, Transfer learning and deep convolutional neural networks for safety guardrail detection in 2D images, *Autom. Constr.* 89 (2018) 58–70, <https://doi.org/10.1016/j.autcon.2018.01.003>.
- [24] W. Fang, L. Ding, B. Zhong, P.E. Love, H. Luo, Automated detection of workers and heavy equipment on construction sites: a convolutional neural network approach, *Adv. Eng. Inform.* 37 (2018) 139–149, <https://doi.org/10.1016/j.aei.2018.05.003>.
- [25] B.E. Mneymneh, M. Abbas, H. Khoury, Vision-based framework for intelligent monitoring of hardhat wearing on construction sites, *J. Comput. Civ. Eng.* 33 (2) (2018) 04018066, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000813](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000813).
- [26] K.K. Han, M. Golparvar-Fard, Appearance-based material classification for monitoring of operation-level construction progress using 4D BIM and site photologs, *Autom. Constr.* 53 (2015) 44–57, <https://doi.org/10.1016/j.autcon.2015.02.007>.
- [27] C. Kropp, C. Koch, M. König, Interior construction state recognition with 4D BIM registered image sequences, *Autom. Constr.* 86 (2018) 11–32, <https://doi.org/10.1016/j.autcon.2017.10.027>.
- [28] H. Hamledari, B. McCabe, S. Davari, Automated computer vision-based detection of components of under-construction indoor partitions, *Autom. Constr.* 74 (2017) 78–94, <https://doi.org/10.1016/j.autcon.2016.11.009>.
- [29] Y. Turkan, F. Bosché, C. T. Haas, R. Haas, Tracking of secondary and temporary objects in structural concrete work, *Constr. Innov.* 14 (2) (2014) 145–167, <https://doi.org/10.1108/CI-12-2012-0063>.
- [30] J. Gong, C.H. Caldas, Computer vision-based video interpretation model for automated productivity analysis of construction operations, *J. Comput. Civ. Eng.* 24 (3) (2009) 252–263, [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000027](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000027).
- [31] H. Kim, Y. Ham, W. Kim, S. Park, H. Kim, Vision-based nonintrusive context documentation for earthmoving productivity simulation, *Autom. Constr.* 102 (2019) 135–147, <https://doi.org/10.1016/j.autcon.2019.02.006>.
- [32] M. Bügler, A. Borrmann, G. Ogunmakin, P.A. Vela, J. Teizer, Fusion of photogrammetry and video analysis for productivity assessment of earthwork processes, *Comput. Aided Civ. Inf. Eng.* 32 (2) (2017) 107–123, <https://doi.org/10.1111/mice.12235>.
- [33] H. Kim, S. Bang, H. Jeong, Y. Ham, H. Kim, Analyzing context and productivity of tunnel earthmoving processes using imaging and simulation, *Autom. Constr.* 92 (2018) 188–198, <https://doi.org/10.1016/j.autcon.2018.04.002>.
- [34] J. Kim, S. Chi, Action recognition of earthmoving excavators based on sequential pattern analysis of visual features and operation cycles, *Autom. Constr.* 104 (2019) 255–264, <https://doi.org/10.1016/j.autcon.2019.03.025>.
- [35] K. Barnard, P. Duygulu, D. Forsyth, N.D. Freitas, D.M. Blei, M.I. Jordan, Matching words and pictures, *J. Mach. Learn. Res.* 3 (2003) 1107–1135, <https://doi.org/10.1109/83.826781>.
- [36] E. Chang, K. Goh, G. Sychay, G. Wu, CBSA: content-based soft annotation for multimodal image retrieval using Bayes point machines, *IEEE Trans. Circuits Syst. Video Technol.* 13 (1) (2003) 26–38, <https://doi.org/10.1109/TCSVT.2002.808079>.
- [37] S. Feng, R. Manmatha, V. Lavrenko, Multiple Bernoulli relevance models for image and video annotation, *Proceedings of the 2004 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR 2004)*, Washington, DC, vol. 2, pp. II-II, DOI, <https://doi.org/10.1109/CVPR.2004.1315274>.
- [38] J. Li, J.Z. Wang, Automatic linguistic indexing of pictures by a statistical modeling approach, *IEEE Trans. Pattern Anal. Mach. Intell.* 25 (9) (2003) 1075–1088, <https://doi.org/10.1109/TPAMI.2003.1227984>.
- [39] J. Mao, W. Xu, Y. Yang, J. Wang, A.L. Yuille, Explain Images with Multimodal Recurrent Neural Networks, *arXiv preprint arXiv:1410.1090*, (2014).
- [40] J.-Y. Pan, H.-J. Yang, P. Duygulu, C. Faloutsos, Automatic image captioning, *Proceedings of the 2004 IEEE International Conference on Multimedia and Expo (ICME 2004)*, Taipei, Taiwan, vol. 3, pp. 1987–1990, <https://doi.org/10.1109/ICME.2004.1394652>.
- [41] S. Ren, K. He, R. Girshick, J. Sun, Faster R-CNN: Towards Real-time Object Detection With Region Proposal Networks, *arXiv preprint arXiv:1506.01497* (2015).
- [42] A. Dutta, A. Zisserman, The VGG Image Annotator (VIA), *arXiv preprint arXiv:1904.10699*, (2019).
- [43] R. Vedantam, C. Lawrence Zitnick, D. Parikh, CIDEr: consensus-based image description evaluation, *Proceedings of the 2015 IEEE conference on computer vision and pattern recognition (CVPR 2015)*, Boston, MA, pp. 4566–4575, <https://doi.org/10.1109/CVPR.2015.7299087>.